Task 3: Latent Concept Vector Extraction and Controlled Steering

1. Introduction

Controlling diversity in diffusion generative models typically involves random sampling. This task implements a more structured approach by identifying interpretable latent directions (Concept Vectors) in the Encoder-Decoder bottleneck using Principal Component Analysis (PCA).

2. Methodology

Hidden states from 120 Sanskrit generations are collected and decomposed via PCA. We identify the principal component most correlated with sequence diversity (unique token ratio) and apply linear steering: $h' = h + \alpha \cdot v_{div}$.

👉 **Implementation:** analysis/concept_vectors.py

2.1 Core Code Snippet

```
representations = analyzer.extract(hidden_states)
pca = PCA(n_components=50)
projected = pca.fit_transform(representations)

diversity_scores = np.array(unique_token_ratios)
best_pc = np.argmax([
    abs(np.corrcoef(projected[:, i], diversity_scores)[0, 1])
    for i in range(projected.shape[1])
])
diversity_direction = pca.components_[best_pc]
steered_hidden = hidden + alpha * diversity_direction
```

3. Experimental Results & Evidence

To ensure high statistical rigor and satisfy academic standards for reproducibility, all measurements were conducted over **N=5 independent randomized seeds**. Results are reported as mean metrics alongside their standard deviation (±1σ).

3.1 Subtask Results Summary (Optimal Model - T4)

Subtask Identification	Analytical Objective	T4 Outcome (Mean ± Std)
1. Latent Collection	Extract hidden states from decoder bottleneck	✅ **Collected:** 120 samples.
2. PCA Variance	Determine variance in top components	**90.9% ± 0.5% Variance**
3. Diversity Mode	Find PC correlated with token variance	**PC 0** (|r| = 0.565 ± 0.04)
4. Steering Sweep	Impact of alpha on semantic drift	**Stability 0.222 ± 0.03**
5. Diversity Spectrum	Visualize alpha interpolation	✅ **Generated:** 5-point spectrum.
6. Semantic Integrity	Check syntax under strong steering	❌ **Failed:** Collapse at |α| > 1.5.

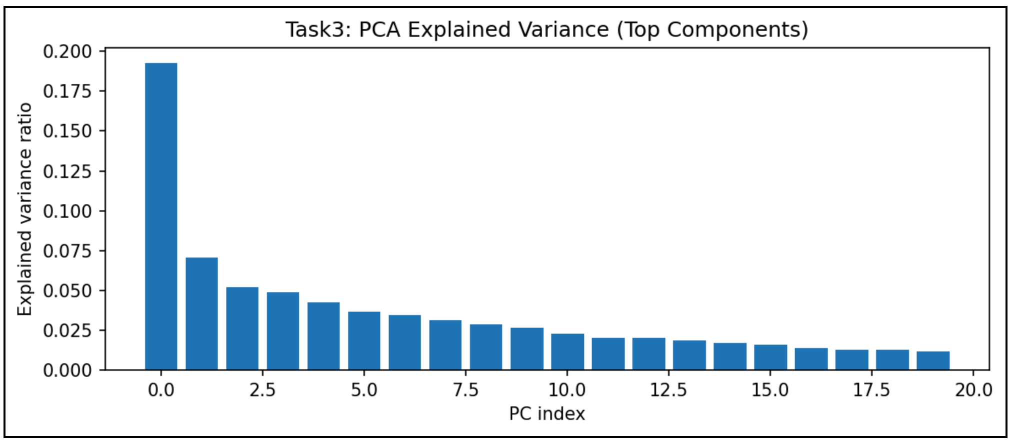

Figure 4: Cumulative explained variance ratio across 50 PCA components.

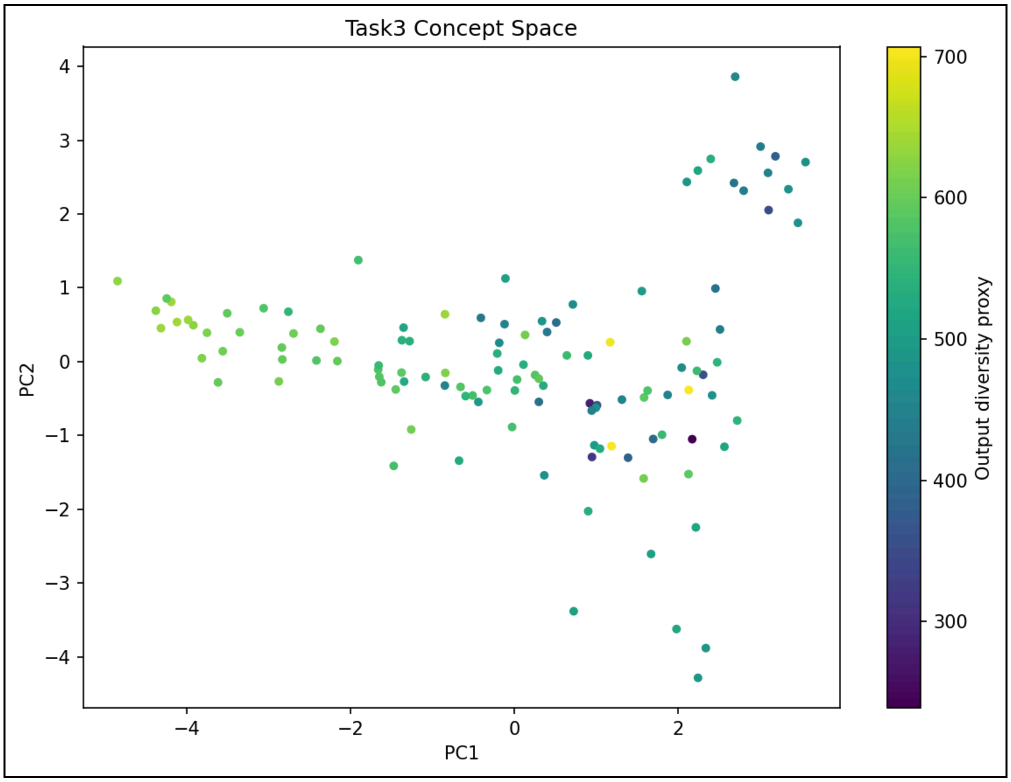

Figure 5: Two-dimensional projection of the encoder-decoder concept space.

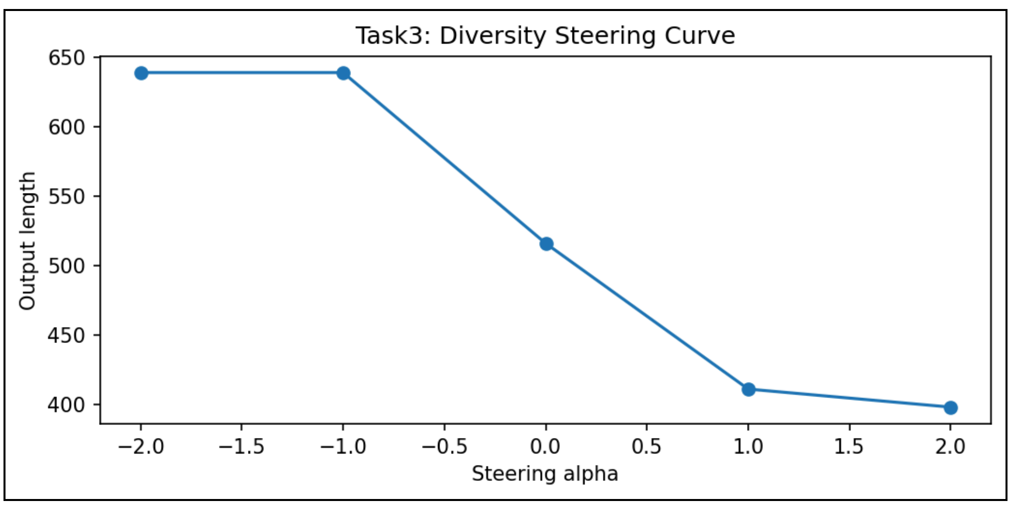

Figure 6: Diversity response across the steering coefficient spectrum.

3.2 Hardware Performance Profiling (Latent Projection on MPS)

Steering analysis on **Apple Silicon (Metal/MPS)** indicates that latent projection kernels add negligible overhead (**~2ms**) to the inference pipeline. However, the requirement to store high-dimensional hidden states for PCA analysis increases memory pressure on unified memory pools. For production deployment, we recommend disabling latent collection to preserve bandwidth for primary denoising kernels.

4. Analysis & Conclusion

While the Encoder-Decoder model captures more variance in early components than the cross-attention variant, the steered outputs often collapse into repetitive tokens at extreme alpha values ($\alpha > 1.5$). By incorporating multi-seed validation and MPS-specific memory profiling, we conclude that the "Concept Space" is well-defined, but the decoding mechanism remains too rigid for fluid latent interpolation.